



BIOLOGY CH.6 — REPRODUCTION IN PLANTS — FULL MIND MAP

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1. Plant Gaseous Exchange & Basic Physiology

- Stomata CO_2 uptake aur oxygen release mein madad karta hai photosynthesis ke dauran - guard cells stomata ke open-close ke liye responsible hain. [Q323]

2. Asexual Reproduction in Plants & Organisms

- Spores asexual reproductive bodies hain jo sporangia mein bante hain - Ferns, Moss, Fungi spore formation se reproduce karte hain. Amoeba binary fission, Yeast budding se reproduce karta hai. [Q324]
- Leishmania → Binary fission, Yeast → Budding, Plasmodium → Multiple fission, Hydra → Regeneration/Budding - ye matching yaad rakho. [Q328]
- Rhizopus (bread mould fungus) spore formation se reproduce karta hai - Phylum Zygomycota, saprophytic/parasitic fungus, moist organic substances par paya jata hai. Sporangium reproductive organ hai, sporangiophore vegetative organ hai. [Q337, Q338, Q392, Q384]
- Fission do types ke hote hain: Binary fission (2 mein divide, e.g., Amoeba) aur Multiple fission (kai daughter cells, e.g., Plasmodium). [Q382]
- Planaria true regeneration dikhata hai - ek piece se poora organism grow ho sakta hai. Spirogyra fragmentation se reproduce karta hai. [Q385]

3. Vegetative Propagation - Natural & Artificial

- Vegetative propagation asexual method hai jisme plants genetically IDENTICAL hote hain parents ke - advantages: faster cultivation, seedless plants grow kar sakte hain; disadvantages: short-lived, diseases ke liye susceptible, no new varieties. [Q329]
- ⚠️ Trap: Vegetative propagation se produced plants genetically 'different' hain - ye galat statement hai, actually identical hote hain. [Q329]

- Sweet potato, Banana, Ginger, Turmeric roots se new plants produce karte hain - Rose stem cutting se, Bryophyllum leaf margins se plantlets grow karta hai. Underground stem modifications: rhizomes, tubers, corms, bulbs. [Q325]
- Mango typically sexually reproduce karta hai seed se ya grafting se - natural vegetative propagation nahi karta. Papaya bhi seeds se grow hota hai, vegetative propagation nahi hota. [Q339, Q374]
- Banana suckers se propagate hota hai - banana ka underground stem rhizome kehlata hai, sucker lateral shoot hota hai base se. [Q370]
- Sugarcane (stem cuttings), Potato (stem tubers with 'eyes'), Banana (rhizome suckers) vegetatively propagate hote hain - Papaya aur Coriander seeds se grow hote hain. [Q410]
- Jasmine generally vegetatively reproduce karta hai (cuttings, suckers, grafting) - iske flowers mein fertile pollen/ovules ki kami hoti hai, seed produce karne ki capacity lost kar chuka hai. [Q395]
- Natural vegetative propagation Mustard mein NAHI hoti - Mustard sexually seeds se reproduce karta hai. Sweet potato/dahlia root tubers se, ginger/potato rhizomes se propagate hote hain. [Q377]
- Plant body ka koi part new individual mein develop ho jaaye usse 'Vegetative propagation' kehte hain - ye asexual method hai. Sexual propagation seeds se multiplication hota hai. [Q372]

4. Seed Structure, Germination & Development

- Dicot seed mein: Plumule → future shoot, Cotyledons → food store, Radicle → future root. Monocot seeds mein single embryonic leaf hoti hai, dicot mein do cotyledons hoti hain jo food store karte hain. Examples: Dicot - Peas, Almonds, Cashews. [Q327]
- Radicle germination ke dauran root system mein develop hota hai - Plumule shoot (stem, leaves) mein, Cotyledon seedling ke liye food reserve hota hai, Epicotyl cotyledons ke upar plumule ke neeche ka hissa hota hai. [Q366]
- Litchi single-seeded fruit hai - Apple, Pomegranate, Papaya multi-seeded hain. Seeds mein embryo, endosperm, seed coat hote hain. Monocot seeds: Paddy, Wheat; Dicot seeds: Chickpea, Litchi, Mango, Coffee. [Q368]
- Seed mein future plant embryo ki form mein hota hai - fertilization ke baad embryo seed ke andar develop hota hai. [Q373]

5. Flower Structure - Male & Female Parts

- Stamens (androecium) male reproductive parts hain - Filament (stalk) + Anther (pollen produce karta hai). Carpel/Pistil (gynoecium) female part hai - Stigma (sticky top, pollen trap), Style (tube), Ovary (ovules contain karti hai). [Q326, Q336, Q369, Q375, Q397, Q399, Q401, Q413, Q407]
- ⚠ Trap: Style male reproductive part NAHI hai - ye female organ Pistil ka part hai. Male parts sirf Anther aur Filament hain jo Stamen banate hain. [Q341]
- ⚠ Trap: Carpel (Pistil) anther ka part NAHI hai - ye female reproductive part hai. Saffron Crocus sativus ke stigma se derive hota hai. [Q387]
- Flower ke 4 main whorls: Calyx (sepals, outermost green), Corolla (petals), Androecium (stamens), Gynoecium/Carpels (centre mein). Carpel female reproductive organ hai. [Q401, Q413]

- Essential reproductive organs: Stamens aur Carpels - Sepals aur Petals accessory/non-essential hain jo protection aur pollinators attract karne mein help karte hain, gametes directly produce nahi karte. [Q404, Q407]

6. Pollination - Types & Agents

- Pollination pollen grains ka anther se stigma tak transfer hai - Self-pollination same flower mein, Cross-pollination different flowers mein hota hai. Agents: birds, bees, water, wind, butterflies. [Q390]
- Entomophily (insect pollination) mein flowers colorful, fragrant, nectar-rich hote hain - Bees aur butterflies most common biotic agents hain. [Q365]
- Wind-pollinated flowers ka pollen light aur spongy (non-sticky) hota hai taaki air currents se easily travel kar sake - in flowers mein large feathery stigmas hote hain drifting pollen pakadne ke liye. [Q400]
- ⚠ Trap: Plants pollination agents NAHI hain - agents air, water, insects, animals, birds hain; plants khud subject hain, agent nahi. [Q402]
- Pollen grains anthers ke andar meiosis se produce hote hain - Locule anther ki internal cavity hai jahan pollen develop hota hai. [Q391, Q412]

7. Fertilization, Post-Fertilization Changes & Fruit Development

- Nishchekan ke baad: Zygote → Embryo, Ovules → Seed, Ovary → Fruit. [Q332, Q344, Q405]
- ⚠ Trap: Flowering plants spore formation se reproduce NAHI karte - ye fungi mein hota hai. Flowering plants sexually (fertilization, pollination) ya asexually (vegetative propagation, budding, tissue culture) reproduce karte hain. [Q355]
- Double fertilization sirf Angiosperms mein hota hai - ek sperm egg se fuse karke embryo banata hai, doosra polar nuclei se fuse karke endosperm banata hai. [Q371]
- Pollen grain stigma par land karke pollen tube grow karta hai male gametes ko ovule tak transfer karne ke liye - Fertilization types: Porogamy (micropyle se), Chalazogamy (chalaza se), Mesogamy (integuments se). [Q350]
- Fertilization ke baad petals, stamens, stigma, style usually dry up aur fall off kar jaate hain - Ovary fruit mein develop hota hai. Sepal defensive part hai jo kuch plants mein persist karta hai (jaise tomato). [Q353]
- Parthenocarpy mein fertilization ke bina fruit develop hota hai (seedless) - examples: Pineapple, Banana, Grape, certain oranges. [Q408]
- Pollen tube ka development Spermatophytes (seed-producing plants) mein land adaptation ka important stage hai - ye male gametes ko directly ovule tak le jaata hai bina external water ke. [Q398]

8. Mendel's Experiments - Laws & Ratios

- Gregor Johann Mendel (Father of Genetics) ne Pisum sativum (garden pea) par 1856-1863 tak experiments kiye - 14 true-breeding varieties select ki. Success ka main reason: ek time mein ek hi character ka study karna. [Q358, Q361]

- Mendel ne pehle cross-fertilization kiya (opposite traits wale plants ke beech), phir F1 progeny ko self-fertilize kiya F2 generation ke liye. [Q331]
- Mendel ne F2 generation F1 hybrids ko self-pollination karke obtain ki. [Q379]
- Mendel ke 7 contrasting traits: Flower colour (Purple/White), Seed shape (Round/Wrinkled), Seed colour (Green/Yellow), Pod shape (Constricted/Inflated), Pod colour (Green/Yellow), Plant size (Tall/Dwarf), Flower position (Axial/Terminal). [Q346, Q342]
- ⚠ Trap: Mendel ne leaf shape ka use NAHI kiya - unhone 7 traits use kiye: flower colour, position, stem length, seed shape, seed colour, pod shape, pod colour. [Q342]
- Dominant traits pea plants mein: Round seeds, Tall height, Purple flowers, Axial position. Recessive traits: Wrinkled seeds, Small (dwarf) height, White flowers, Terminal position. [Q330]
- F2 dihybrid cross ka phenotypic ratio 9:3:3:1 hota hai - 9 round-yellow : 3 round-green : 3 wrinkled-yellow : 1 wrinkled-green. [Q333, Q360]
- Standard monohybrid cross F2 generation ka genotypic ratio 1:2:1 hota hai (1 TT : 2 Tt : 1 tt), phenotypic ratio 3:1 hota hai (3 tall : 1 dwarf). [Q334, Q349]
- TT × tt cross mein F1 generation all tall hoti hai kyunki tallness dominant hai - chahe pollen tall plant se ho ya short se, result same aata hai. [Q335, Q357, Q359]
- ⚠ Trap: F2 generation mein 'all plants tall' ye statement INCORRECT hai - F2 mein 3:1 ratio mein tall aur dwarf dono hote hain. [Q356]
- 3:1 ratio mein sirf ek trait dominant hota hai, doosra recessive - dono dominant nahi hote (red dominant, green recessive). [Q340]
- Agar Mendel F2 generation study NAHI karte toh unhe pata nahi chalta ki recessive traits lost nahi hote, bas suppressed hote hain aur baad mein reappear karte hain. [Q363]
- RRyy (round, green) × rrYY (wrinkled, yellow) → F1 RrYy (round, yellow) hota hai kyunki R aur Y dono dominant hain. [Q347]
- TtWw genotype 4 types ke gametes produce karta hai: TW, Tw, tW, tw - independent assortment se har type 25% hota hai. [Q345]
- 500 peas F1 self-pollination se collect karne par green peas ≈125 honge (25% of 500 based on 3:1 ratio). [Q354]
- Tallness ko dominant isliye kaha gaya kyunki plant tall dikhta hai chahe ek (Tt) ya do (TT) alleles ho - phenotype mein sirf dominant trait expressed hoti hai. [Q348]
- Tall wrinkled (TTrr) × Short round (ttRR) cross se F1 generation TtRr hogi jo Tall round dikhegi - Tallness aur Roundness dono dominant traits hain. [Q362]
- Mendel ke experiments se pata chalta hai ki germ cells har pair se ek chromosome leti hain - Law of Segregation ke through DNA stability maintain hoti hai. [Q351]
- ⚠ Trap: 'Two alleles gamete formation ke time together laaye jaate hain' - ye FALSE statement hai. Law of Segregation ke anusaar alleles gamete formation ke dauran SEPARATE hote hain. [Q352]
- Monohybrid cross mein sirf ek pair of contrasting characters involved hoti hai, dihybrid cross mein do pairs involved hoti hain. [Q388, Q389]
- F1 generation mein dominant trait clearly dikhai deti hai (effective/intensity) - contrasting traits F1 ya F2 mein kabhi blend nahi hote. [Q380, Q381, Q386]

9. Plant Classification - Angiosperms, Gymnosperms, Cryptogams

- Angiosperms (flowering plants) mein seeds ovary ke andar develop hote hain jo pakkar fruit banta hai - reproductive organs flower mein hote hain. [Q367, Q378, Q393, Q394]
- Gymnosperms mein naked, exposed seeds hote hain (Cycads, Ginkgo, Pinus) - ye cones ya modified leaves par hote hain, fruit mein enclosed nahi hote. [Q414]
- Thallophyta primitive plants hain (Spirogyra, Ulva), Pteridophyta flowerless hain (Fern, Selaginella). [Q367]
- Marsilea Cryptogams se belong karta hai - iske reproductive organs hidden hote hain (spores se reproduce, no flowers/seeds). [Q364]

10. Tissue Culture & Modern Techniques

- Tissue culture (micropropagation) mein isolated cells/tissues ko artificial nutrient medium par grow kiya jata hai - isse ek parent se kai disease-free plants grow kar sakte hain. Gottlieb Haberlandt ne discover kiya. [Q383, Q411]

11. Plant Tissues & Meristem

- Meristematic tissue cell division capable hai - small cells indefinitely divide kar sakti hain, growing regions (root/stem apex) mein payi jaati hai. Sclerenchyma dead structural support cells, Parenchyma soft parts, Xylem complex permanent tissue. [Q406]

12. Monocots vs Dicots

- Dicots mein 2 cotyledons, tap root system, branched stems, reticulate venation hoti hai - Fibrous roots aur parallel venation monocots ki characteristics hain, isliye 'dicots have fibrous roots' galat statement hai. [Q403]

13. Bisexual vs Unisexual Flowers

- Unisexual flowers mein sirf male ya female parts hote hain - Examples: Papaya, Watermelon, Pumpkin, Pine, Coconut, Cucumber, Maize, Birch. [Q343, Q376]
- Bisexual (hermaphrodite) flowers mein stamen aur pistil dono hote hain - Examples: Mustard, Rose, Lily, Tulip, Hibiscus, Sunflower, Tomato, Gulmohar. [Q343, Q396, Q409]

14. Spore Formation & Fungi

- Rhizopus, mushrooms, Aspergillus spores se reproduce karte hain - spores reproductive cells hain jo fusion ke bina new organism develop kar sakti hain. [Q384]

